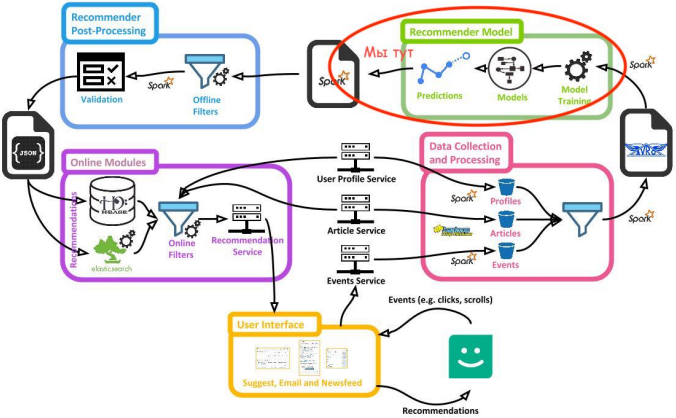
education

Рекомендательные системы

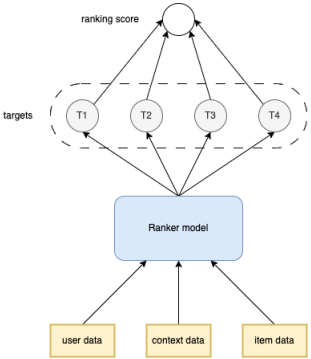
Николай Анохин

Контекст


Recommender Post-Processing
Validation
Spark
Offline Filters
Spark
Мы тут
Recommender Model
Predictions
Models
Model Training
Spark
JSON
Online Modules
Recommendations
elasticsearch
Online Filters
Recommendation Service
User Profile Service
Article Service
Events Service
Data Collection and Processing
Spark
Profiles
Articles
Events
Spark
Spark
Events (e.g. clicks, scrolls)
User Interface
Suggest, Email and Newsfeed
Recommendations

Истории успеха: ранжирование

Постановка задачи ранжирования



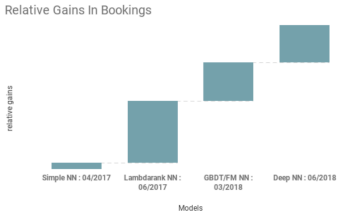
Задача
Выбрать порядок кандидатов так, чтобы самые релевантные стояли в голове списка.

Истории успеха: ранжирование

Как оставить след в науке

- Победить xgboost
- Пофиксить смещения

Applying Deep Learning To Airbnb Search [HAR+19]



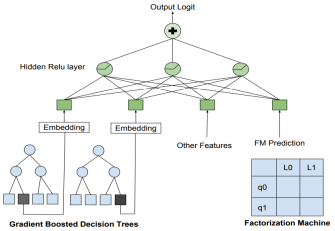
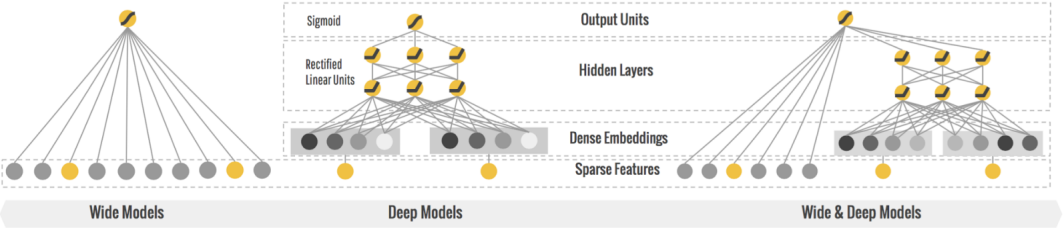


Figure 3: NN with GBDT tree nodes and FM prediction as features

> ...we were able to deprecate all that complexity by simply scaling the training data 10x and moving to a DNN with 2 hidden layers...

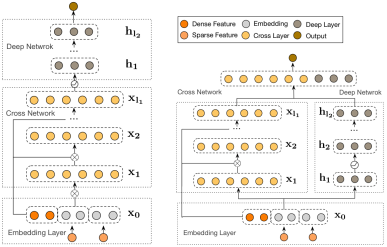
Идея	Специфика применения NN ранкера на практике
Интересность	★★★★
Полезность	★★★★★

Wide & Deep Learning: Better Together with TensorFlow [Che16]



Идея	Скомбинировать “ручные” перемножения признаков с нелинейностью
Интересность	★★
Полезность	★★★

DCN V2: Improved Deep & Cross Network and Practical Lessons for Web-scale Learning to Rank Systems [WSC+21]



(a) Stacked (b) Parallel

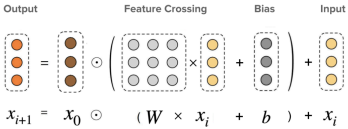
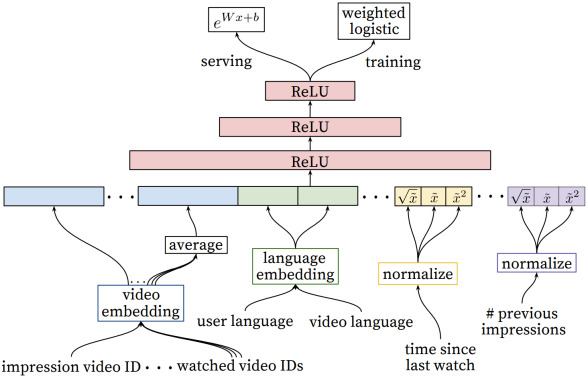


Figure 2: Visualization of a cross layer.

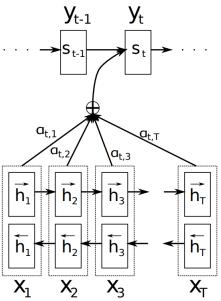
Идея	Интегрировать перемножения признаков в архитектуру NN
Интересность	★★★
Полезность	★★★★

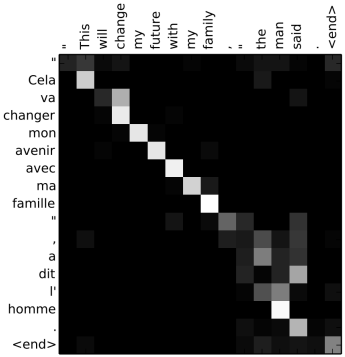
YouTube: ранжирование



Идея	Использовать последовательность айтемов как признак
Интересность	★★★★★
Полезность	★★★★★

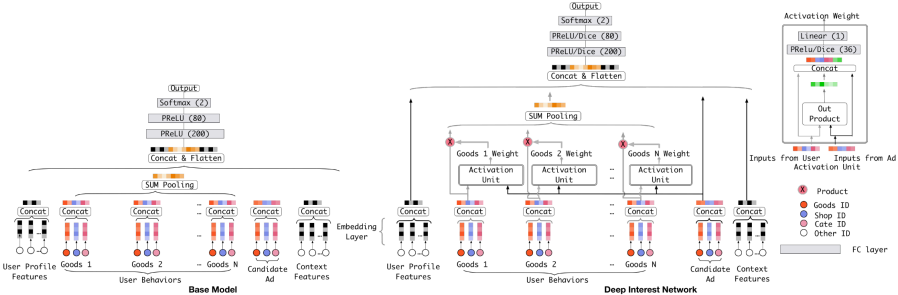
Механизм attention [BCB16]





$$\alpha_{ij} = \frac{\exp e_{ij}}{\sum_k \exp e_{ik}}, \quad e_{ij} = \mathrm{ffn}(s_{i-1}, h_j)$$

Deep Interest Network for Click-Through Rate Prediction [ZSZ+18]



Идея	Attention (почти) для агрегации последовательности айтемов
Интересность	★★★★
Полезность	★★★★

MaskNet: Introducing Feature-Wise Multiplication to CTR Ranking Models by Instance-Guided Mask [WSZ2

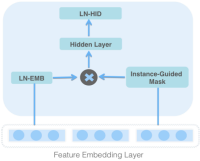


Figure 2: MaskBlock on Feature Embedding

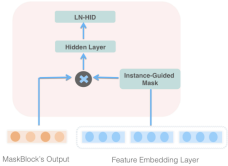


Figure 3: MaskBlock on MaskBlock

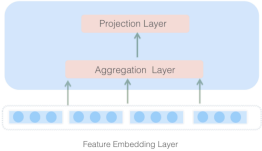


Figure 1: Neural Structure of Instance-Guided Mask

MaskNet: Introducing Feature-Wise Multiplication to CTR Ranking Models by Instance-Guided Mask [WSZ2

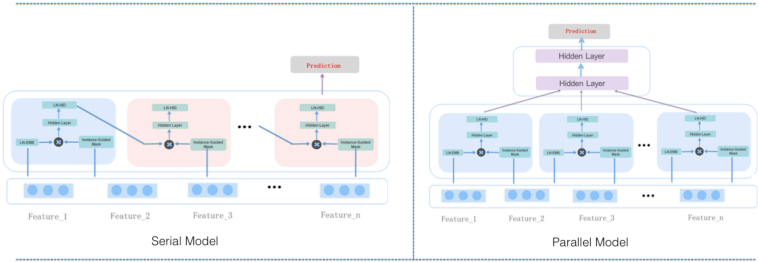
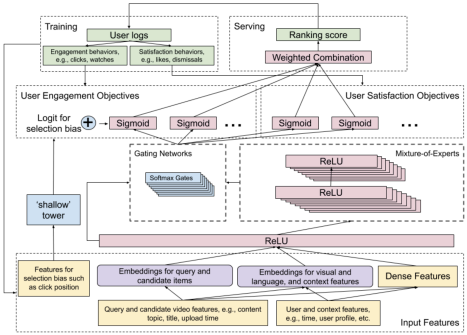


Figure 4: Structure of Serial Model and Parallel Model

Идея	Обучаемая маска выбирает “полезные” части выходов слоя
Интересность	⋆⋆⋆⋆
Полезность	⋆⋆⋆⋆

Recommending What Video to Watch Next: A Multitask Ranking System [ZHW+19]



Идея	Attention выбирает экспертов под разные таргеты
Интересность	★★★★★
Полезность	★★★★

Истории успеха: контент

Истории успеха: контент

Как оставить след в науке

- Решить проблему холодного старта, хитро обучив эмбединги

Learning Transferable Visual Models From Natural Language Supervision [RKH+21]

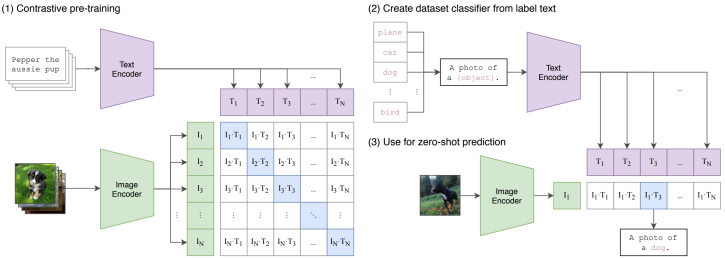
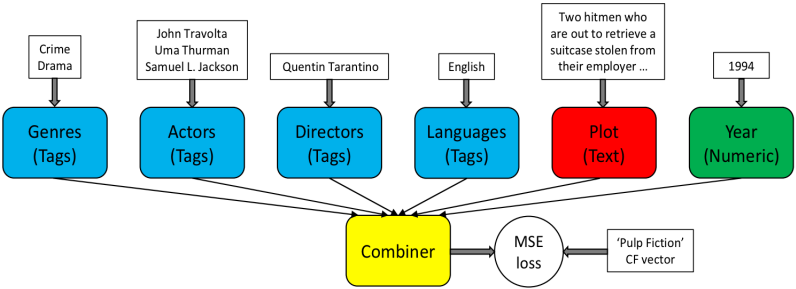


Figure 1. Summary of our approach. While standard image models jointly train an image feature extractor and a linear classifier to predict some label, CLIP jointly trains an image encoder and a text encoder to predict the correct pairings of a batch of (image, text) training examples. At test time the learned text encoder synthesizes a zero-shot linear classifier by embedding the names or descriptions of the target dataset's classes.

Идея	Мульти-модальные эмбеддинги решают вопросы
Интересность	★★★
Полезность	★★★

CB2CF: A Neural Multiview Content-to-Collaborative Filtering Model for Completely Cold Item Recommendations [BKYK19]



Идея	Использум результат коллаборативной модели для обучения контентно
Интересность	★★★★
Полезность	★★★★★

Что работает на практике

Архитектуры [Wan23]

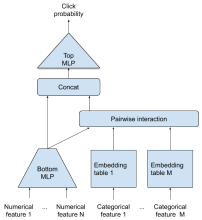
- Мульти-таргет решает.
- Авто пересечение признаков (DCN, MaskNet).
- MoE не дает профита при том же количестве параметров, как у базовых архитектур.
- DIN дает эффект для длинных последовательностей.

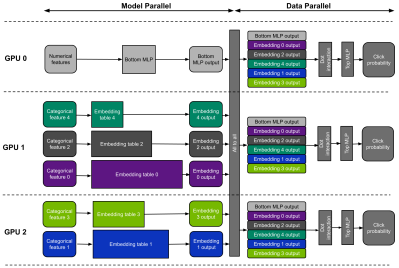
Improving Deep Learning For Airbnb Search [HAR+20]

> If deeper nets were not the right architecture for us, we hypothesized, more specialized architectures might be. So we tried architectures [...] like **deep and wide** [...] followed by variants of **attention based networks**. [...] The short summary of those efforts is that they **failed to move the needle**.

Идея	Архитектура должна следовать из проблемы пользователя
Интересность	⋆⋆⋆⋆⋆
Полезность	⋆⋆⋆⋆⋆

Deep Learning Recommendation Model for Personalization and Recommendation Systems [NMS+19]





Идея	Простая архитектура, но много параметров в эмбеддингах[1]
Интересность	★★★
Полезность	★★★★★

[1] https://catalog.ngc.nvidia.com/orgs/nvidia/resources/dlrm_for_pytorch

Проблемы нейрорекомендеров

Проблема воспроизводимости [DCJ19]

Многие результаты из статей невозможно воспроизвести

Некоторые новые алгоритмы работают хуже, чем затюненные бейзлайны

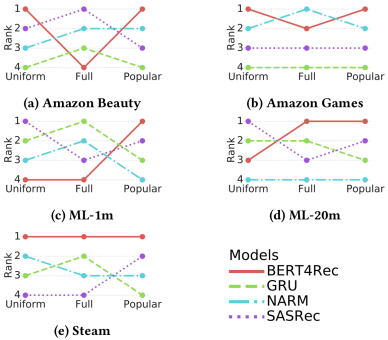
The CMN method was presented at SIGIR 18 and combines memory networks and neural attention mechanisms with latent factor and neighborhood models

	Pinterest			
	HR@5	NDCG@5	HR@10	NDCG@10
TopPopular	0.1668	0.1066	0.2745	0.1411
UserKNN	**0.6886**	**0.4936**	0.8527	**0.5470**
ItemKNN	**0.6966**	**0.4994**	**0.8647**	**0.5542**
$P^3\alpha$	0.6871	**0.4935**	0.8449	**0.5450**
$RP^3\beta$	**0.7018**	**0.5041**	**0.8644**	**0.5571**
CMN	0.6872	0.4883	0.8549	0.5430

	Epinions			
	HR@5	NDCG@5	HR@10	NDCG@10
TopPopular	**0.5429**	**0.4153**	**0.6644**	**0.4547**
UserKNN	0.3506	0.2983	0.3922	0.3117
ItemKNN	0.3821	0.3165	0.4372	0.3343
$P^3\alpha$	0.3510	0.2989	0.3891	0.3112
$RP^3\beta$	0.3511	0.2980	0.3892	0.3103
CMN	0.4195	0.3346	0.4953	0.3592

Проблема сравнений [DZH21]

Результат сравнения может поменяться на обратный в зависимости от того, по какой метрике сравнивают

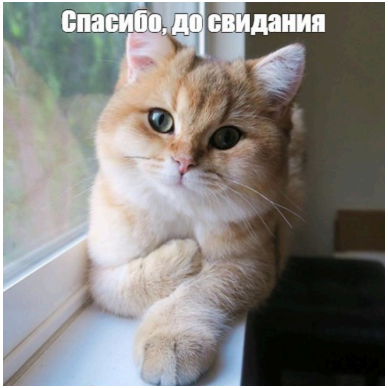


Итоги

Итоги

Нейросетевые модели могут заменить любой компонент рекомендательной системы: отборщик кандидатов, ранкер, item2item.

Нейросетевой подход не гарантирует выигрыша – к выбору модели нужно подходить прагматично.


Спасибо, до свидания

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